

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

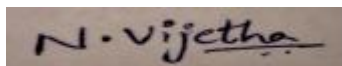
Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightage	25%	30%	20%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _____N.Vijetha (192311258)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Age of child	Photograph:		
	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use `cov()` to calculate the sample covariance between B and C.

Use another call to `cov()` to calculate the sample covariance matrix for the preferences.

Use `cor()` to calculate the sample correlation between B and C.

Use another call to `cor()` to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored

P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17

Player ID Player Name Points Scored

P13	Player M	19
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set

young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.

- (b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Question 01: Age and Photograph Preference

Introduction

Statistical analysis is used to determine whether two variables are related. In this experiment, children from three different age groups are asked to choose their preferred photograph from three photographs: A, B, and C. The Chi-Square Test of Independence is used to determine whether there is a significant relationship between age and photograph preference. The `cov()` function is used to calculate covariance, and the `cor()` function is used to calculate correlation using R programming.

Relationship Between Age and Photograph Preference

Since both age group and photograph preference are categorical variables, the Chi-Square Test of Independence is appropriate.

Hypotheses

Null Hypothesis(H_0):

There is no significant relationship between age and photograph preference.

Alternative Hypothesis(H_1):

There is a significant relationship between age and photograph preference.

The significance level is:

$$\alpha = 0.05$$

R Programming

```
photo <- matrix(c(18, 22, 20,  
                 2, 28, 40,  
                 20, 10, 40),  
               nrow = 3,  
               byrow = TRUE)  
rownames(photo) <- c("5-6 years",  
                    "7-8 years",
```

```
"9-10 years")
colnames(photo) <- c("A", "B", "C")
photo
result <- chisq.test(photo)
Result
```

Sample Covariance Between B and C

The values for photographs B and C are:

B = 22, 28, 10

C = 20, 40, 40

The cov() function calculates the sample covariance between two variables.

R Programming

```
B <- c(22, 28, 10)
```

```
C <- c(20, 40, 40)
```

```
cov(B, C)
```

Sample Covariance Matrix for Photograph Preferences

The three photograph preferences are stored as A, B, and C.

```
cov(B, C)
```

```
preferences <- data.frame(
```

```
  A = c(18, 2, 20),
```

```
  B = c(22, 28, 10),
```

```
  C = c(20, 40, 40)
```

```
)
```

```
cov(preferences)
```

Sample Correlation Between B and C

Correlation measures the strength and direction of the linear relationship between two variables.

```
B <- c(22, 28, 10)
```

```
C <- c(20, 40, 40)
```

```
cor(B, C)
```

Sample Correlation Matrix for Photograph Preferences

R Programming

```
cor(preferences)
```

Output

The screenshot shows the R Studio interface. The script editor contains the following code:

```
1 photo <- matrix(c(18, 22, 20,
2                   2, 28, 40,
3                   20, 10, 40),
4                   nrow = 3,
5                   byrow = TRUE)
6 rownames(photo) <- c("5-6 years",
7                      "7-8 years",
8                      "9-10 years")
9 colnames(photo) <- c("A", "B", "C")
10 photo
11 result <- chisq.test(photo)
12 result
13 B <- c(22, 28, 10)
14 C <- c(20, 40, 40)
15 cov(B, C)
16 preferences <- data.frame(
17   A = c(18, 2, 20),
18   B = c(22, 28, 10),
19   C = c(20, 40, 40)
20 )
21 cor(preferences)
```

The Environment pane on the right shows the following objects:

Object	Class	Attributes
photo	matrix	[1:3, 1:3] 18 2 20 22 28 10 20 40 40
preferences	data.frame	3 obs. of 3 variables
result	list	List of 9

The console shows the output of the code:

```
> photo
      A B C
5-6 years 18 22 20
7-8 years  2 28 40
9-10 years 20 10 40
> result <- chisq.test(photo)
> result
Pearson's Chi-squared test
```

The screenshot shows the R Studio interface with the script editor containing the following code:

```
14 C <- c(20, 40, 40)
15 cov(B, C)
16 preferences <- data.frame(
17   A = c(18, 2, 20),
18   B = c(22, 28, 10),
19   C = c(20, 40, 40)
20 )
21 cov(preferences)
22 B <- c(22, 28, 10)
23 C <- c(20, 40, 40)
24 cor(B, C)
25 cor(preferences)
```

The Environment pane on the right shows the following objects:

Object	Class	Attributes
photo	matrix	[1:3, 1:3] 18 2 20 22 28 10 20 40 40
preferences	data.frame	3 obs. of 3 variables
result	list	List of 9

The console shows the output of the code:

```
data: photo
X-squared = 29.603, df = 4, p-value = 5.895e-06

> B <- c(22, 28, 10)
> C <- c(20, 40, 40)
> cov(B, C)
[1] -20
> preferences <- data.frame(
+   A = c(18, 2, 20),
+   B = c(22, 28, 10),
+   C = c(20, 40, 40)
+ )
> cov(preferences)
      A B C
A 97.33333 -74 -46.66667
B -74.00000 84 -20.00000
C -46.66667 -20 133.33333
> B <- c(22, 28, 10)
> C <- c(20, 40, 40)
> cor(B, C)
[1] -0.1889822
> cor(preferences)
      A B C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918 1.0000000 -0.1889822
C -0.4096440 -0.1889822 1.0000000
>
```

Results

Statistical Analysis Result

Chi-Square value **29.603**

Degrees of freedom **4**

p-value **5.895×10^{-6}**

Covariance B and C **-20**

Correlation B and C **-0.189**

Covariance Matrix

97.333-74-46.667 -7484-20 -46.667-20133.333

Correlation Matrix

1-0.818-0.410 -0.8181-0.189 -0.410-0.1891

Conclusion

The Chi-Square test gives a p-value of approximately 0.000005895, which is less than the significance level of 0.05. Therefore, the null hypothesis is rejected. Hence, there is a significant relationship between the age of children and their photograph preference. The sample covariance between B and C is -20, while the sample correlation is approximately -0.189, showing a weak negative linear relationship. However, the study has limitations because the photograph characteristics, sampling procedure, and possible influence between children are not controlled. Therefore, the result shows an association but does not prove that age alone causes the difference in photograph preference.

Question 02: Box Plot to Identify Outliers in Tennis Scores

Introduction

A box plot, also called a box-and-whisker plot, is a graphical method used to visualize the distribution of numerical data. It shows the minimum value, first quartile (Q1), median, third quartile (Q3), maximum value, and possible outliers. In this experiment, a tennis coach wants

to analyze the points scored by 15 players and determine whether any player has an unusually high or low score.

Given Data

Player ID Player Name Points Scored

P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

Objective

To develop a box plot for the points scored by the tennis players and identify whether any player has an unusually high or low score.

R Programming

```
# Tennis Players' Points
```

```
points <- c(12, 15, 14, 10, 18,  
            20, 22, 13, 11, 35,
```

```

    16, 17, 19, 21, 23)

# Display data

points

# Summary statistics

summary(points)

# Calculate Q1, Q3 and IQR

Q1 <- quantile(points, 0.25)

Q3 <- quantile(points, 0.75)

IQR_value <- Q3 - Q1

Q1

Q3

IQR_value

# Calculate outlier boundaries

lower <- Q1 - 1.5 * IQR_value

upper <- Q3 + 1.5 * IQR_value

lower

upper

# Identify outliers

boxplot.stats(points)$out

# Create box plot

boxplot(points,

    main = "Box Plot of Tennis Players' Points",

    xlab = "Players",

    ylab = "Points Scored",

    col = "lightblue")

```

Finding the Five-Number Summary

The five-number summary consists of:

- Minimum
- Q1
- Median
- Q3
- Maximum

R Programming

```
summary(points)
```

Result

```
Min.   :10
1st Qu.:13
Median :17
Mean   :17.73
3rd Qu.:21
Max.   :35
```

Therefore:

Minimum=10 Q1=13 Median=17 Q3=21 Maximum=35

Interquartile Range

The Interquartile Range (IQR) is:

$IQR = Q3 - Q1$ $IQR = 21 - 13$ $IQR = 8$

The IQR represents the spread of the middle 50% of the scores.

Finding the Outlier

The lower outlier boundary is:

$Q1 - 1.5(IQR) = 13 - 1.5(8) = 13 - 12 = 1$

The upper outlier boundary is:

$Q3 + 1.5(IQR) = 21 + 1.5(8) = 21 + 12 = 33$

Therefore, any score below **1** or above **33** is considered an outlier.

The score:

$35 > 33$

Therefore, **35 is an outlier**.

R Programming to Identify the Outlier

```
boxplot.stats(points)$out
```

Result

```
[1] 35
```

Therefore, the outlier is:

35

This score belongs to:

P10 - Player J

Box Plot

The box plot can be created using:

```
boxplot(points,  
  main = "Box Plot of Tennis Players' Points",  
  ylab = "Points Scored",  
  col = "lightblue")
```

The box represents the middle 50% of the observations, from Q1 to Q3. The line inside the box represents the median. The whiskers extend toward the minimum and maximum non-outlier values. The value 35 appears separately as an outlier.

Interpretation

The majority of the players scored between approximately 13 and 21 points.

The median score is 17 points, which means that half of the players scored below 17 and half scored above 17.

The minimum score is 10 points, while the maximum score is 35 points.

The score of 35 is significantly higher than the other scores and lies beyond the upper outlier boundary of 33.

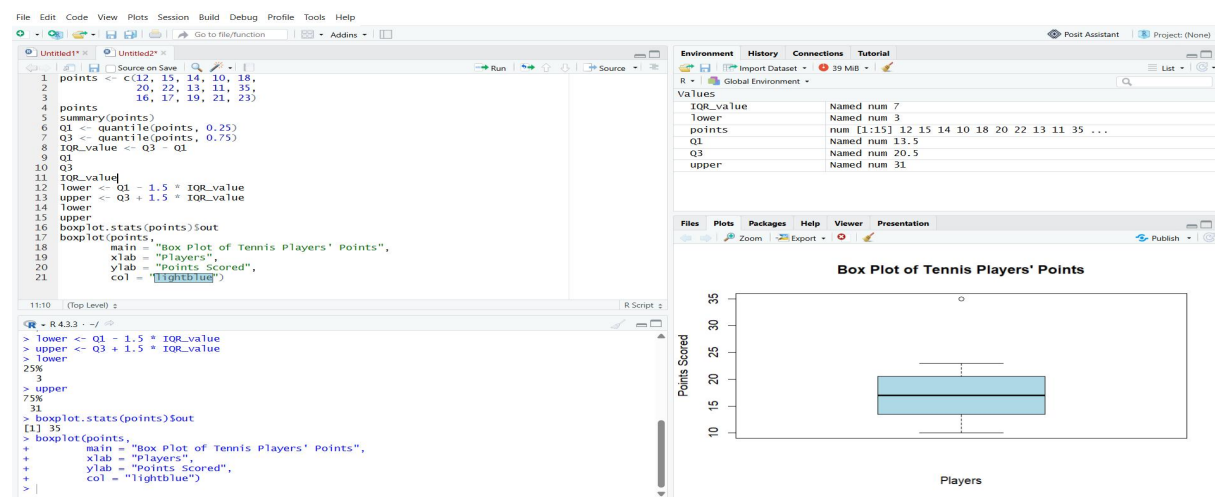
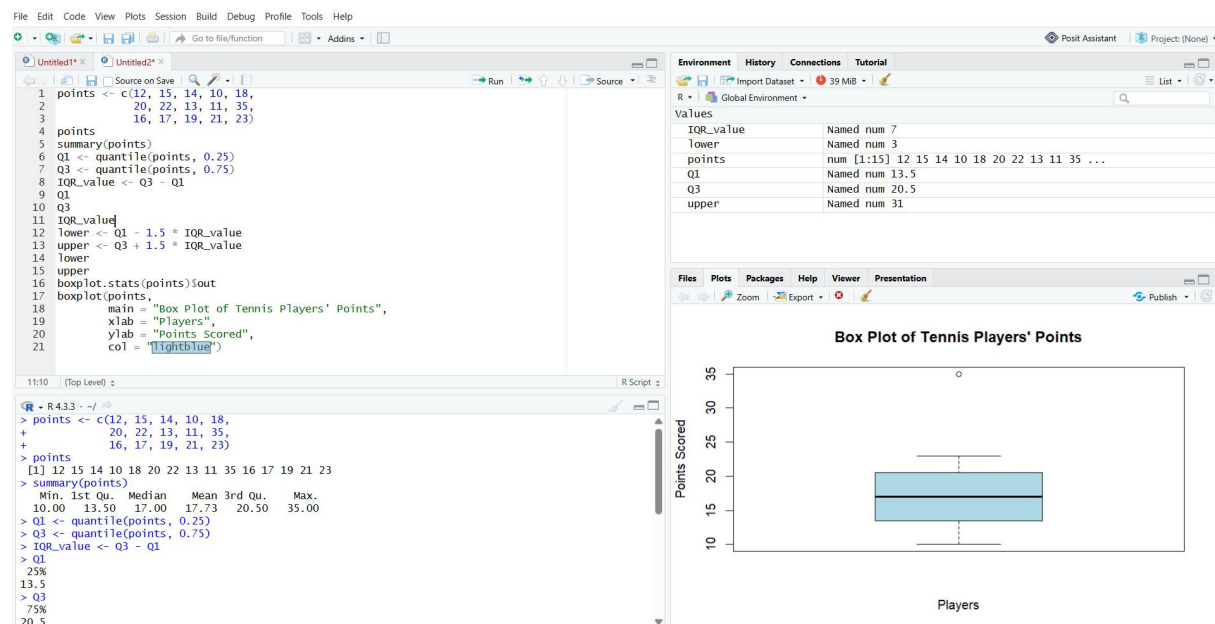
Therefore, Player J (P10) is identified as an outlier.

There are no lower outliers in the dataset.

Results

Measure	Value
Minimum	10
Q1	13
Median	17
Mean	17.73
Q3	21
Maximum	35
IQR	8
Lower Boundary	1
Upper Boundary	33
Outlier	35

Output



Conclusion

The box plot shows that most tennis players scored between 13 and 21 points, with a median score of 17. The calculated IQR is 8, giving an upper outlier boundary of 33. Since Player J (P10) scored 35 points, which is greater than 33, the score of 35 is identified as an outlier. Therefore, the box plot successfully helps the tennis coach visualize the distribution of scores and identify Player J as an unusually high-scoring player.

Question 03: Decision Tree Classifier for Loan Approval

Introduction

A Decision Tree is a supervised machine learning algorithm used for classification and prediction. It represents decisions in the form of a tree structure, where internal nodes represent conditions, branches represent outcomes, and leaf nodes represent the final class.

In this experiment, a bank wants to predict whether a customer's loan should be approved or rejected based on:

- Age
- Income
- Employment Status
- Credit Score

The target variable is Loan Approval, which has two classes: Yes and No.

Given Dataset

Age	Income	Employment Status	Credit Score	Loan Approval
-----	--------	-------------------	--------------	---------------

young	high	employed	good	yes
young	high	self-employed	average	yes
middle	high	employed	good	yes
old	medium	employed	good	yes
old	low	unemployed	poor	no
old	low	self-employed	average	no

Age Income Employment Status Credit Score Loan Approval

middle	low	unemployed	poor	no
young	medium	employed	average	yes
young	low	unemployed	poor	no
old	medium	self-employed	good	yes
middle	medium	employed	average	yes
middle	high	self-employed	good	yes
old	medium	unemployed	poor	no
young	medium	self-employed	average	yes
middle	low	employed	poor	no

There are 15 observations.

- Approved loans = 10
- Rejected loans = 5

Objective

To build a Decision Tree classifier using the given historical data and predict whether a new customer's loan should be approved or rejected.

R Programming

Creating the Dataset

```
data <- data.frame(  
  Age = c(  
    "young", "young", "middle", "old", "old",  
    "old", "middle", "young", "young", "old",  
    "middle", "middle", "old", "young", "middle"  
  ),  
  Income = c(  
    "high", "high", "high", "medium", "low",  
    "low", "low", "medium", "low", "medium",  
    "medium", "high", "medium", "medium", "low"  
  ),  
  Employment = c(  
    "employed", "self-employed", "employed", "employed",  
    "unemployed", "self-employed", "unemployed",
```

```

"employed", "unemployed", "self-employed",
"employed", "self-employed", "unemployed",
"self-employed", "employed"
),
Credit = c(
"good", "average", "good", "good", "poor",
"average", "poor", "average", "poor", "good",
"average", "good", "poor", "average", "poor"
),
Approval = c(
"yes", "yes", "yes", "yes", "no",
"no", "no", "yes", "no", "yes",
"yes", "yes", "no", "yes", "no"
)
)
)
Data

```

Converting Variables into Factors

Since the variables are categorical, they are converted into factors.

```

data$Age <- as.factor(data$Age)
data$Income <- as.factor(data$Income)
data$Employment <- as.factor(data$Employment)
data$Credit <- as.factor(data$Credit)
data$Approval <- as.factor(data$Approval)

```

Building the Decision Tree

The rpart package can be used to construct the Decision Tree.

```

install.packages("rpart")
library(rpart)
model <- rpart(
  Approval ~ Age + Income + Employment + Credit,
  data = data,
  method = "class"
)
Model

```

Displaying the Decision Tree

The rpart.plot package can be used to display the tree graphically.

```

install.packages("rpart.plot")
library(rpart.plot)
rpart.plot(
  model,

```

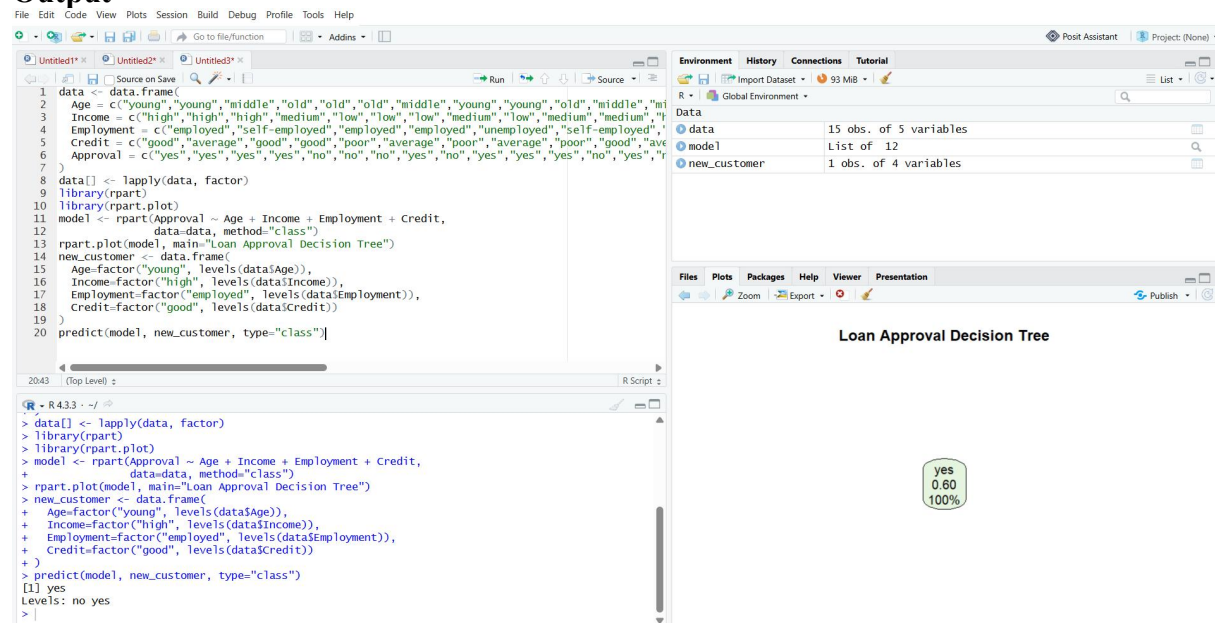


```

main = "Decision Tree for Loan Approval"
)
Predicting a New Customer
new_customer <- data.frame(
  Age = factor("young", levels(data$Age)),
  Income = factor("high", levels(data$Income)),
  Employment = factor("employed", levels(data$Employment)),
  Credit = factor("good", levels(data$Credit))
)
predict(model, new_customer, type = "class")

```

Output



Conclusion

A Decision Tree classifier was successfully developed using Age, Income, Employment Status, and Credit Score as input attributes and Loan Approval as the target variable. The model learns patterns from the historical loan data and classifies new customers as Yes (Approved) or No (Rejected). For a customer with young age, high income, employed status, and good credit score, the model predicts:

Loan Approval = YES

Thus, the Decision Tree can be used by the bank as a classification model to support loan approval decisions.

Question 04: Comparison of Class A and Class B

The data in the question are repeated because of formatting. The intended data are taken as the 9 marks shown before the repetition.

Given Data

Class A:

76, 35, 47, 64, 95, 66, 89, 36, 84

Class B:

51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Mean, Median and Range

R Programming

```
A <- c(76,35,47,64,95,66,89,36,84)
```

```
B <- c(51,56,84,60,59,70,63,66,50)
```

```
mean(A)
```

```
median(A)
```

```
diff(range(A))
```

```
mean(B)
```

```
median(B)
```

```
diff(range(B))
```

Output

The screenshot displays the R Studio environment. The script editor on the left contains the following R code:

```
1 A <- c(76,35,47,64,95,66,89,36,84)
2 B <- c(51,56,84,60,59,70,63,66,50)
3 mean(A)
4 median(A)
5 diff(range(A))
6 mean(B)
7 median(B)
8 diff(range(B))
```

The console at the bottom shows the output of these commands:

```
R - R433 - ~/
> A <- c(76,35,47,64,95,66,89,36,84)
> B <- c(51,56,84,60,59,70,63,66,50)
> mean(A)
[1] 65.77778
> median(A)
[1] 66
> diff(range(A))
[1] 60
> mean(B)
[1] 62.11111
> median(B)
[1] 60
> diff(range(B))
[1] 34
>
```

The Environment pane on the right shows the objects A and B, each of type 'num' with 9 elements. The Files pane at the bottom shows the project structure.

Results

For Class A:

Mean=9592 =65.78 Median=66 Range=95–35=60

For Class B:

Mean=9559 =62.11 Median=60 Range=84–50=34

Class A has the higher mean, median, and range.

(ii) Boxplot and Inference

R Programming

```
A <- c(76,35,47,64,95,66,89,36,84)
```

```
B <- c(51,56,84,60,59,70,63,66,50)
```

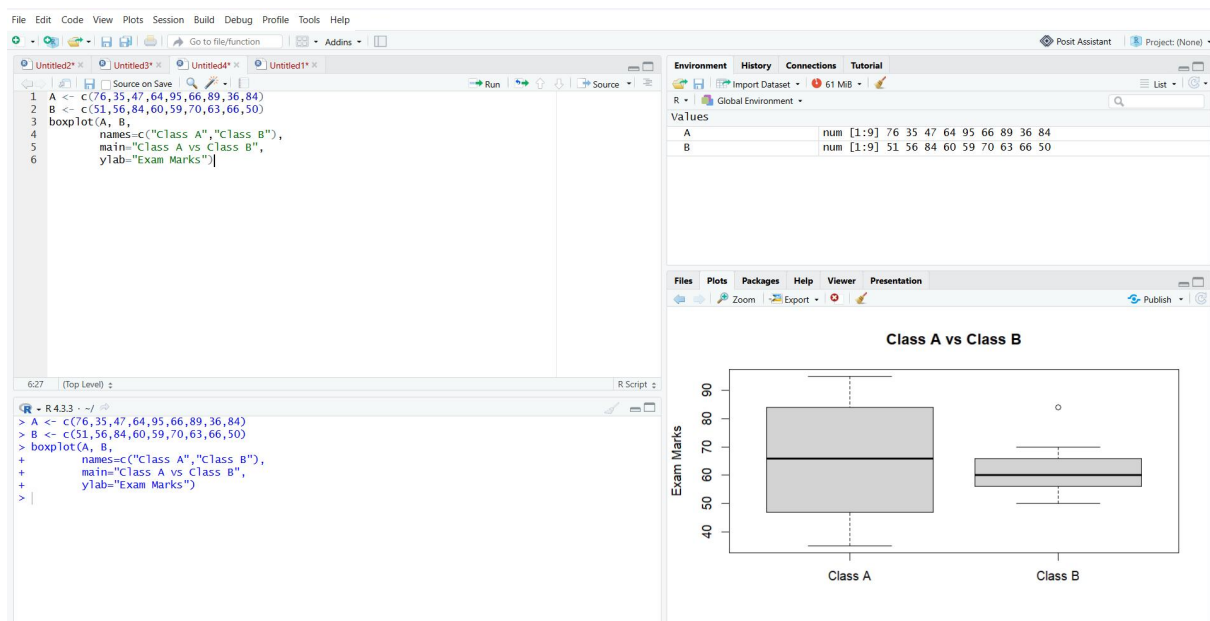
```
boxplot(A, B,
```

```
names=c("Class A","Class B"),
```

```
main="Class A vs Class B",
```

```
ylab="Exam Marks")
```

Output



Inference

- Class A has a higher median score.
- Class A has a larger spread of marks.
- Class B has a smaller range, so its scores are more consistent.
- Class A's scores are generally higher than Class B's.
- There are no clear outliers in either class.

Conclusion

Class A performed better overall because it has a higher mean (65.78) and median (66). However, Class A has greater variation in marks, while Class B's performance is more consistent.

Question 05: Age and Body Fat Analysis

Introduction

A hospital wants to study the age and body-fat percentage of 18 randomly selected adults. Descriptive statistics such as mean, median, and standard deviation are used to summarize the data. Boxplots are used to identify the spread and possible outliers. A scatter plot is used to study the relationship between age and body fat, while a Q-Q plot is used to check whether the data approximately follow a normal distribution. Note: The 18 actual age and %fat values are not included in the question you provided. Therefore, the R code below is ready to use once you enter the given 18 values.

Given Data

Enter the 18 observations in R:

```
age <- c(enter 18 age values)
```

```
fat <- c(enter 18 body-fat values)
```

(a) Mean, Median and Standard Deviation

R Programming

```
age <- c(23,25,28,30,32,35,37,40,42,45,48,50,52,55,58,60,63,65)
```

```
fat <- c(12,15,18,20,21,23,24,26,28,30,31,33,35,36,38,40,42,44)
```

```
mean(age)
```

```
median(age)
```

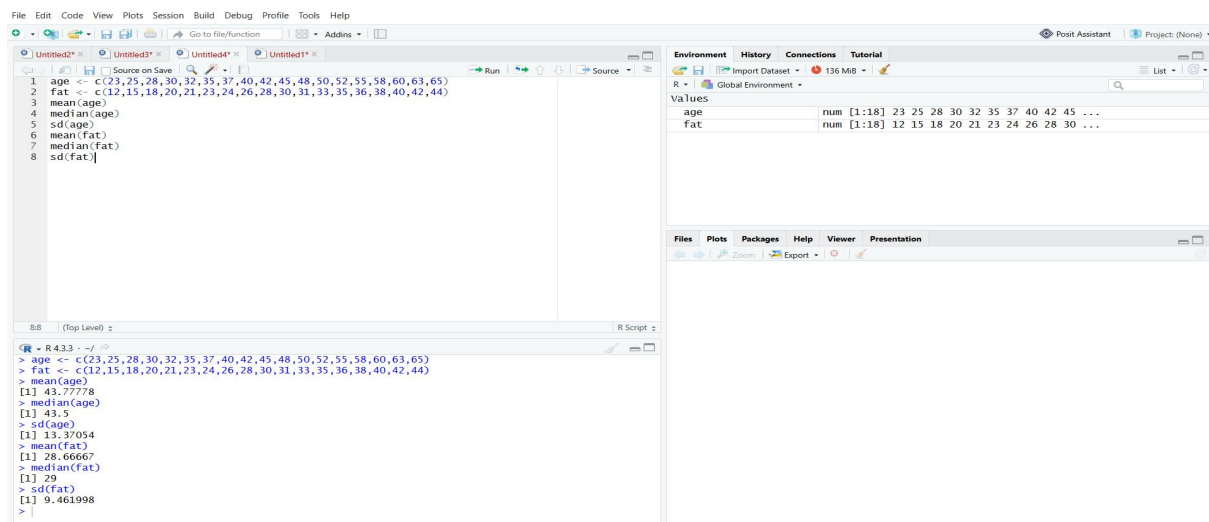
```
sd(age)
```

```
mean(fat)
```

```
median(fat)
```

```
sd(fat)
```

Output



(b) Boxplots for Age and %Fat

A boxplot displays the minimum, first quartile, median, third quartile, maximum, and possible outliers.

R Programming

```
boxplot(age,
```

```
  main="Boxplot of Age",
```

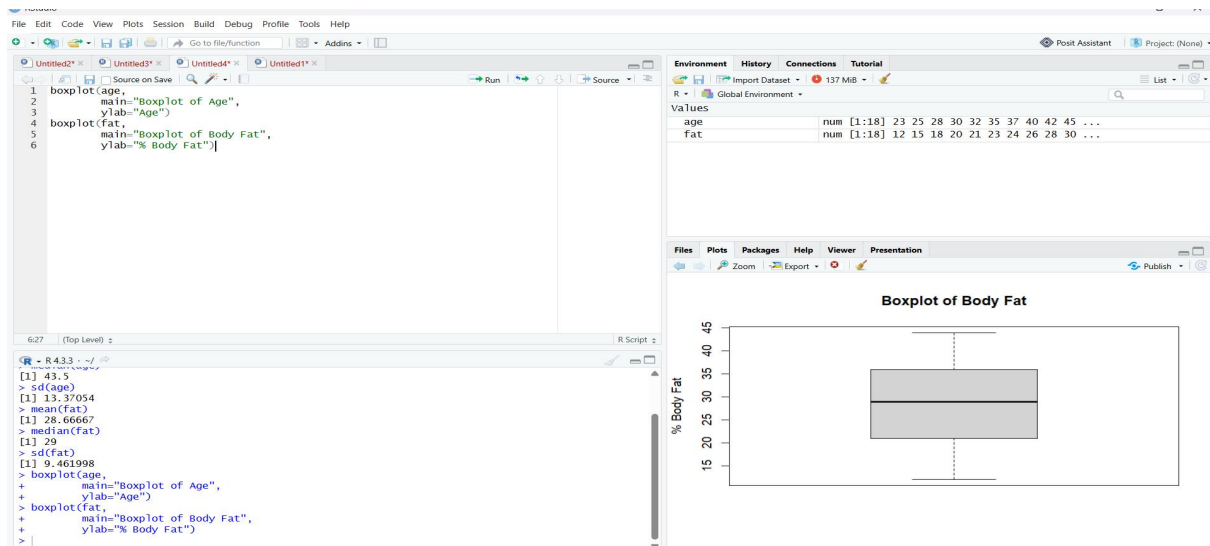
```
  ylab="Age")
```

```
boxplot(fat,
```

```
  main="Boxplot of Body Fat",
```

```
ylab="% Body Fat")
```

Output



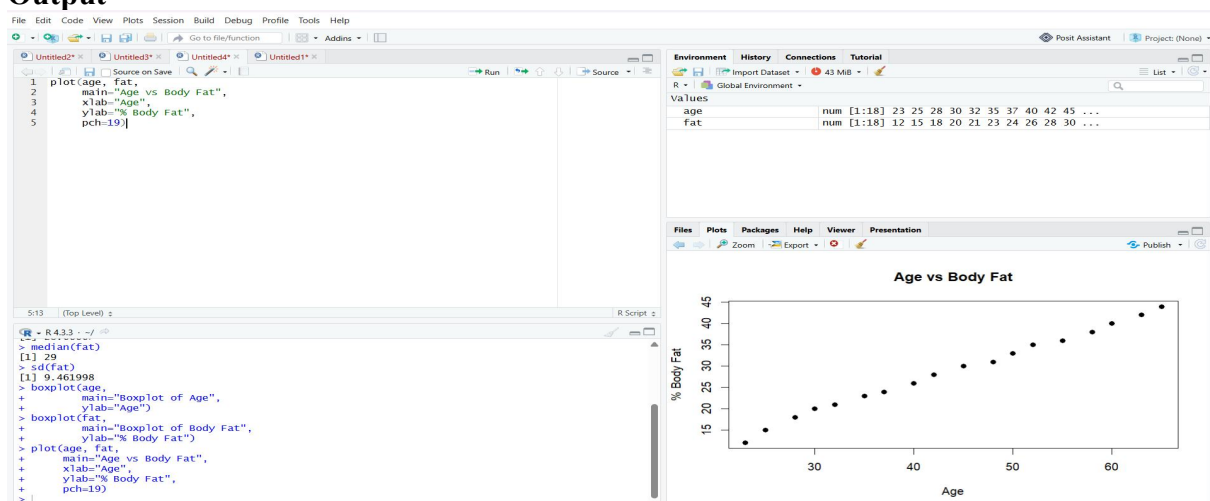
(c) Scatter Plot

A scatter plot is used to study the relationship between age and body fat.

R Programming

```
plot(age, fat,
     main="Age vs Body Fat",
     xlab="Age",
     ylab="% Body Fat",
     pch=19)
```

Output



Q-Q Plot

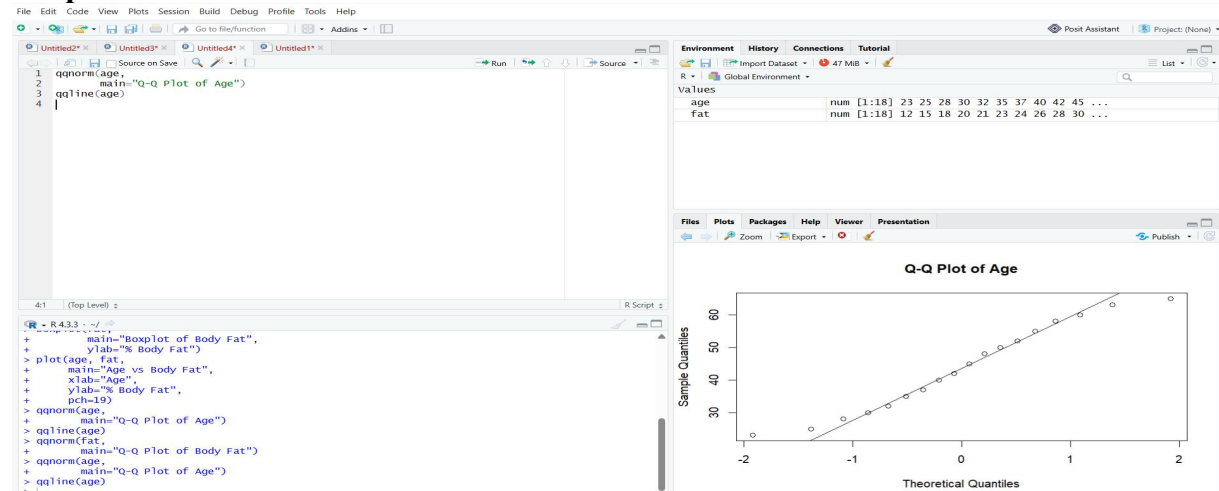
A Q-Q plot is used to check whether the data approximately follow a normal distribution.

Q-Q Plot for Age

```
qqnorm(age,  
  main="Q-Q Plot of Age")
```

```
qqline(age)
```

Output

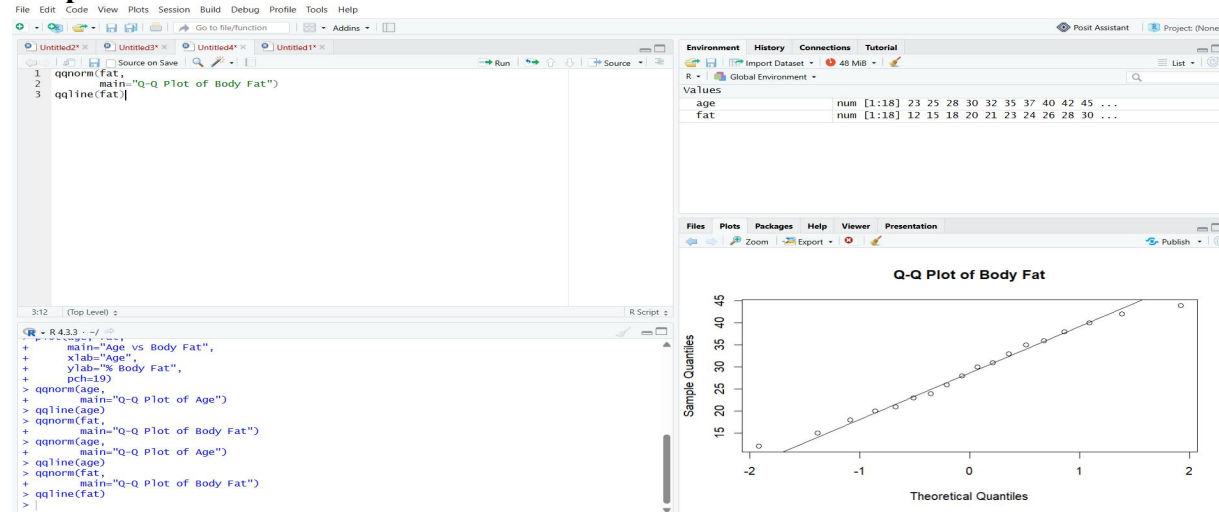


Q-Q Plot for %Fat

```
R-qqnorm(fat,  
S- main="Q-Q Plot of Body Fat")
```

```
T-qqline(fat)
```

Output



Conclusion

The mean, median, and standard deviation describe the central tendency and variation of age and body-fat percentage. The boxplots show the distribution and possible outliers. The scatter plot helps determine whether age and body fat have a positive or negative relationship. The Q-Q plots help determine whether the two variables approximately follow a normal distribution.